📘 Pandas — Interview Notes

Quick one-line definitions • Python / Data Science / AIML

• Creating & Reading Data

pd.Series()	→ Creates a one-dimensional labeled array.
pd.DataFrame()	→ Creates a two-dimensional table with rows and columns.
pd.read_csv()	→ Reads data from a CSV file into a DataFrame.
pd.read_excel()	→ Reads data from an Excel file into a DataFrame.
df.to_csv()	→ Exports a DataFrame to a CSV file.

• Inspecting Data

df.head()	→ Returns the first 5 rows by default.
df.tail()	→ Returns the last 5 rows by default.
df.shape	→ Returns the number of rows and columns as (rows, columns).
df.columns	→ Returns the names of all columns.
df.index	→ Returns the row index labels.
df.info()	→ Shows columns, data types, non-null values, and memory usage.
df.describe()	→ Provides a statistical summary of numerical columns.
df.dtypes	→ Returns the data type of each column.

• Selecting Data

df["column"]	→ Selects a single column.
df[["A","B"]]	→ Selects multiple columns.
df.loc[]	→ Selects rows and columns using labels.
df.iloc[]	→ Selects rows and columns using integer positions.
df.isnull()	→ Detects missing values.

• Missing Values & Cleaning

df.isnull().sum()	→ Counts missing values in each column.
df.dropna()	→ Removes rows or columns containing missing values.
df.fillna()	→ Replaces missing values with specified values.
df.drop()	→ Removes specified rows or columns.
df.rename()	→ Renames rows or column labels.
df.sort_values()	→ Sorts data using one or more columns.

• Analysis & Aggregation

df.value_counts()	→ Counts the frequency of unique values.
df.groupby()	→ Groups data for analysis.
df.agg()	→ Applies one or more aggregation functions.
df.apply()	→ Applies a function to values, rows, or columns.
df.merge()	→ Combines DataFrames using common keys.
pd.concat()	→ Combines multiple DataFrames along rows or columns.

df.join() → *Combines DataFrames using indexes by default.*

df.duplicated() → *Identifies duplicate rows.*

• *String & Date Functions*

df.drop_duplicates() → *Removes duplicate rows.*

df["column"].str.lower() → *Converts strings to lowercase.*

df["column"].str.upper() → *Converts strings to uppercase.*

df["column"].str.strip() → *Removes leading and trailing spaces.*

pd.to_datetime() → *Converts values to Pandas datetime format.*

df["date"].dt.year → *Extracts the year from a datetime column.*

• *Basic Statistics*

df["column"].mean() → *Calculates the average.*

df["column"].sum() → *Calculates the total.*

df["column"].max() → *Returns the maximum value.*

df["column"].min() → *Returns the minimum value.*

df["column"].median() → *Returns the middle value.*

df["column"].count() → *Counts non-null values.*

• □ *Top 10 to Memorize First*

read_csv → head → info → describe → loc → iloc

isnull → fillna → groupby → merge

Tip: Understand each function and practice one small example.